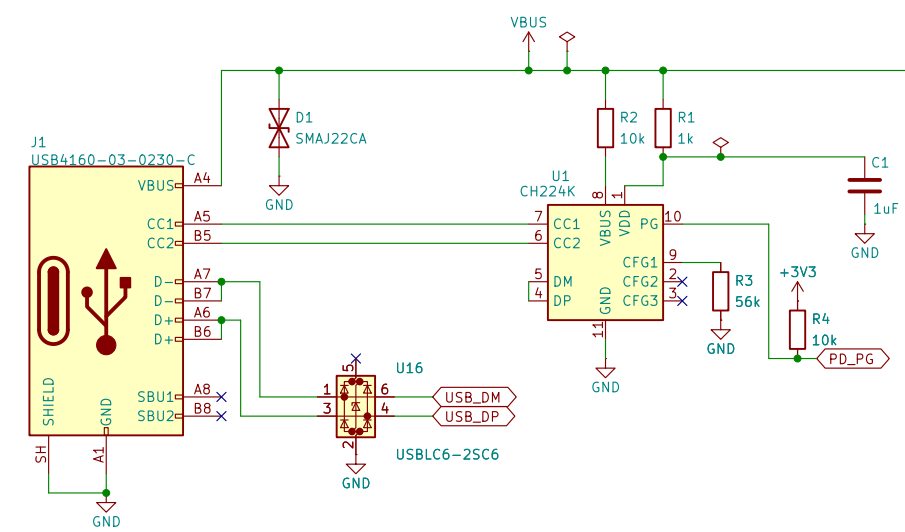
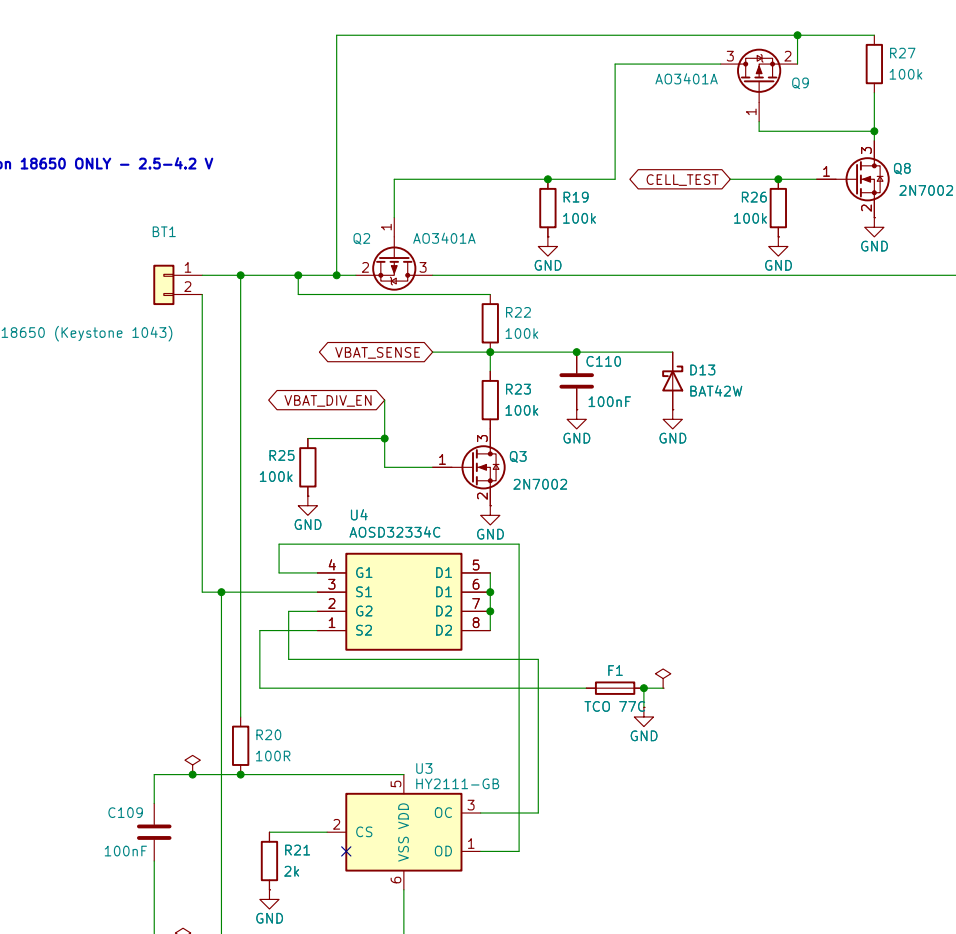


## POWER IN – USB-C PD sink (15 V)



CH224K DP/DM shorted = PD-only; Type-C D+ go to the MCU.  
CFG1=56k -> 15 V P0D; falls back to 5 V (charger idles <11.2 V).  
USB 2.0 subvert of the 24-pin USB4160 (SS pads unconnected).  
D+/D- = native USB-JTAG (019/20). Vertical - port out the back wall.

## BATTERY – 18650 + reverse P-FET + protector + TCO

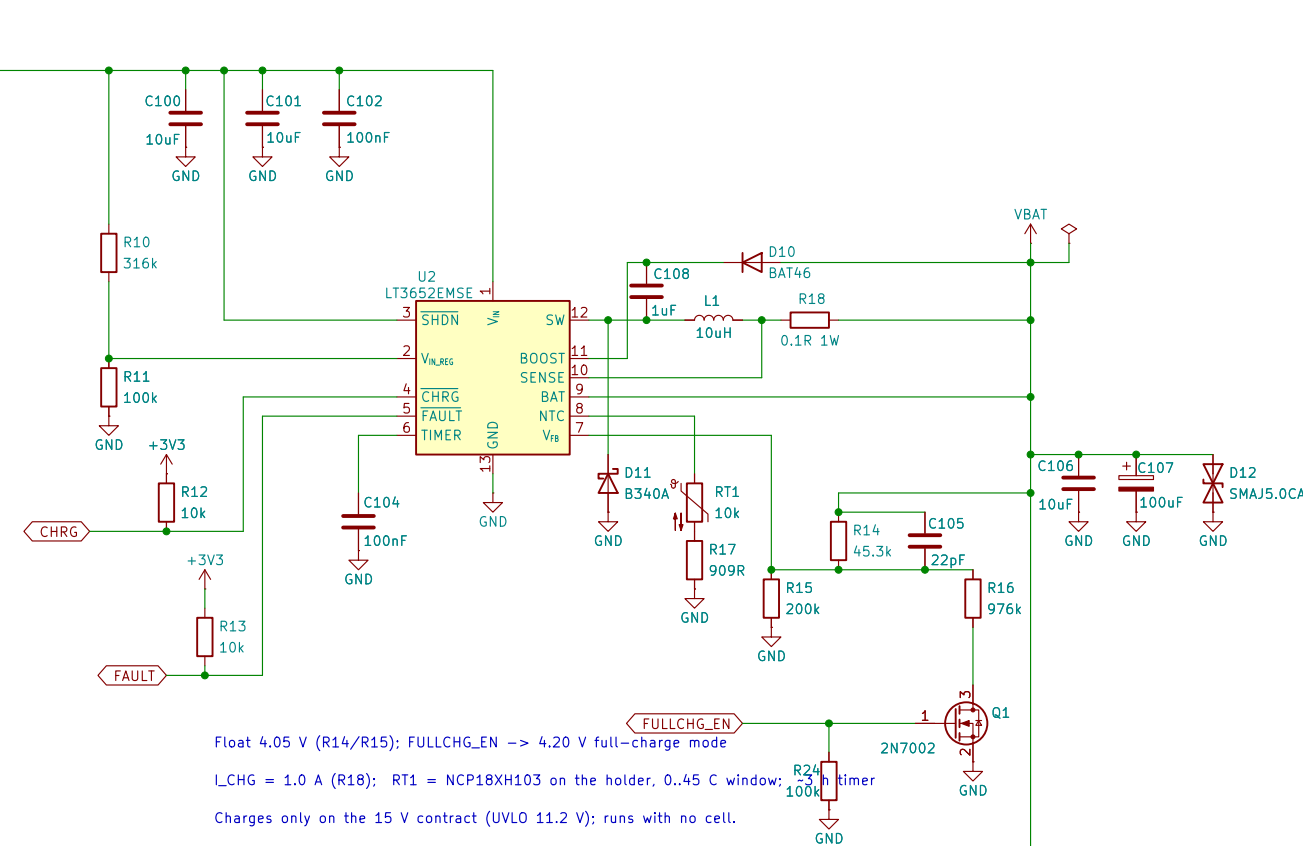


cell- -> A0503233AC (00/0C gated) -> TCO 77 C -> PACK- (GND)  
0V 4.28 V / 0D 2.90 V fixed (-G8). Independent of the charger.

## HOW TO READ THIS SHEET

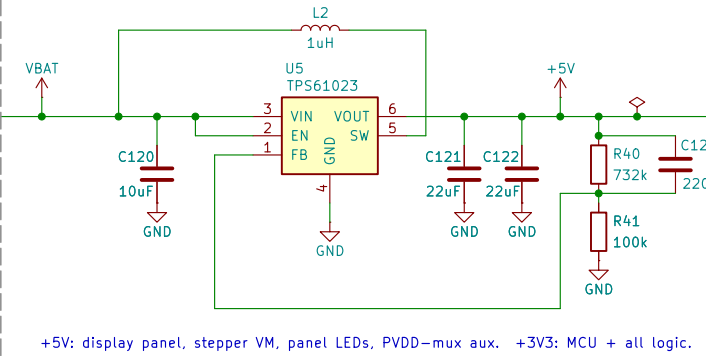
One page, laid out like block\_diagram.drawio: power flows left.  
MCU centre, peripherals right. Power paths and MCU buses (SPI, I2S, MCPWM, encoder) are drawn as wires; GND / +3V3 / +5V taps and named labels connect by name (same name = same net).  
Off-board parts (SD card, sensors, speaker, wake-LED strips, knob, radio toggle, cell) enter via connectors J1/J11 / BT1.

## CHARGER – LT3652 (15 buck, 4.05 V float, 1.0 A)



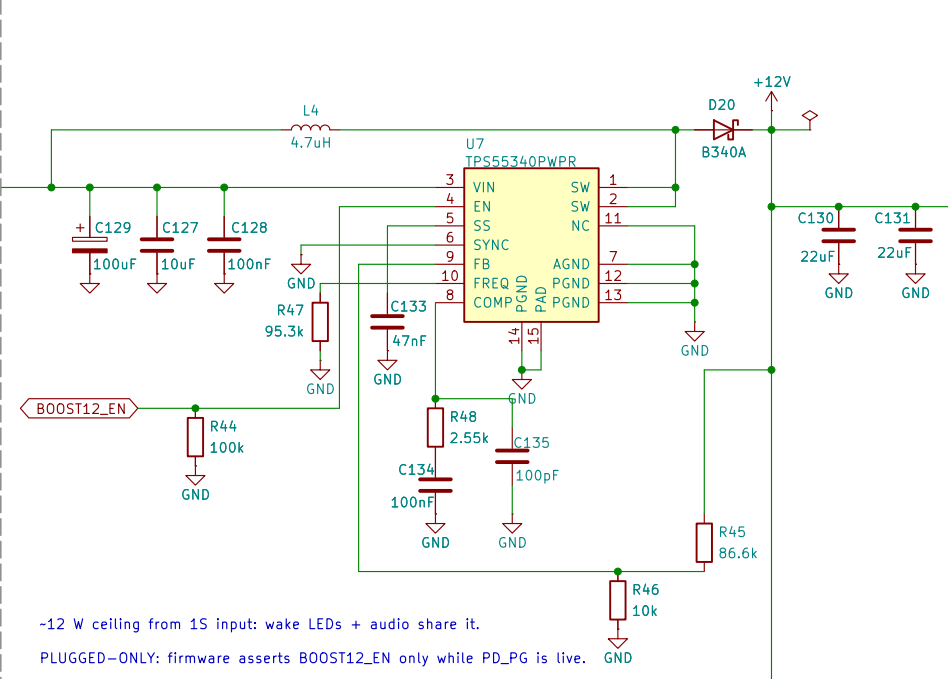
Float 4.05 V (R14/R15): FULLCHG\_EN -> 4.20 V full-charge mode  
L\_CHG = 1.0 A (R18); RT1 = MCP18X103 on the holder. 0.45 C window.  
Charges only on the 15 V contract (UWL0 11.2 V); runs with no coil.

## RAILS – 5 V boost (always on) -> 3V3 buck



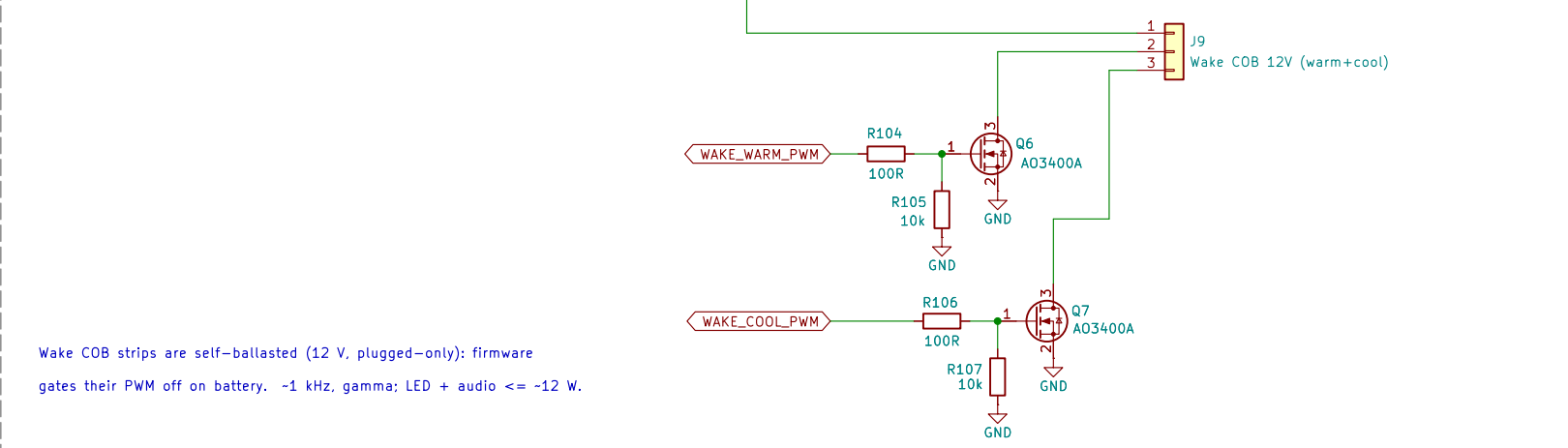
+5V: display panel, stepper VM, panel LEDs, PVDD-mux aux. +3V3: MCU + all logic.

## RAIL – 12 V boost (plugged-only: audio PVDD + wake LEDs)



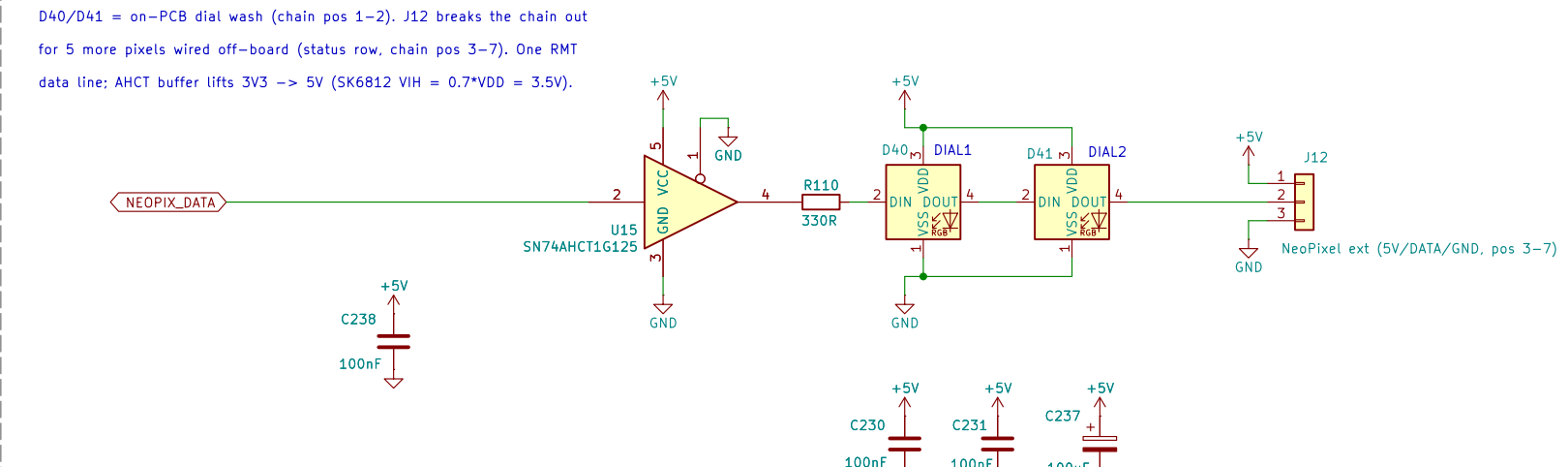
-12 V ceiling from 15 input: wake LEDs = audio share it.  
PLUGGED-ONLY: firmware asserts BOOST12\_EN only while PD\_PG is live.

## LED – wake COB drivers (12 V, plugged-only)



Wake COB strips are self-ballasted (12 V, plugged-only); firmware gates their PWM off on battery. ~1 kHz, gamma: LED = audio <= -12 W.

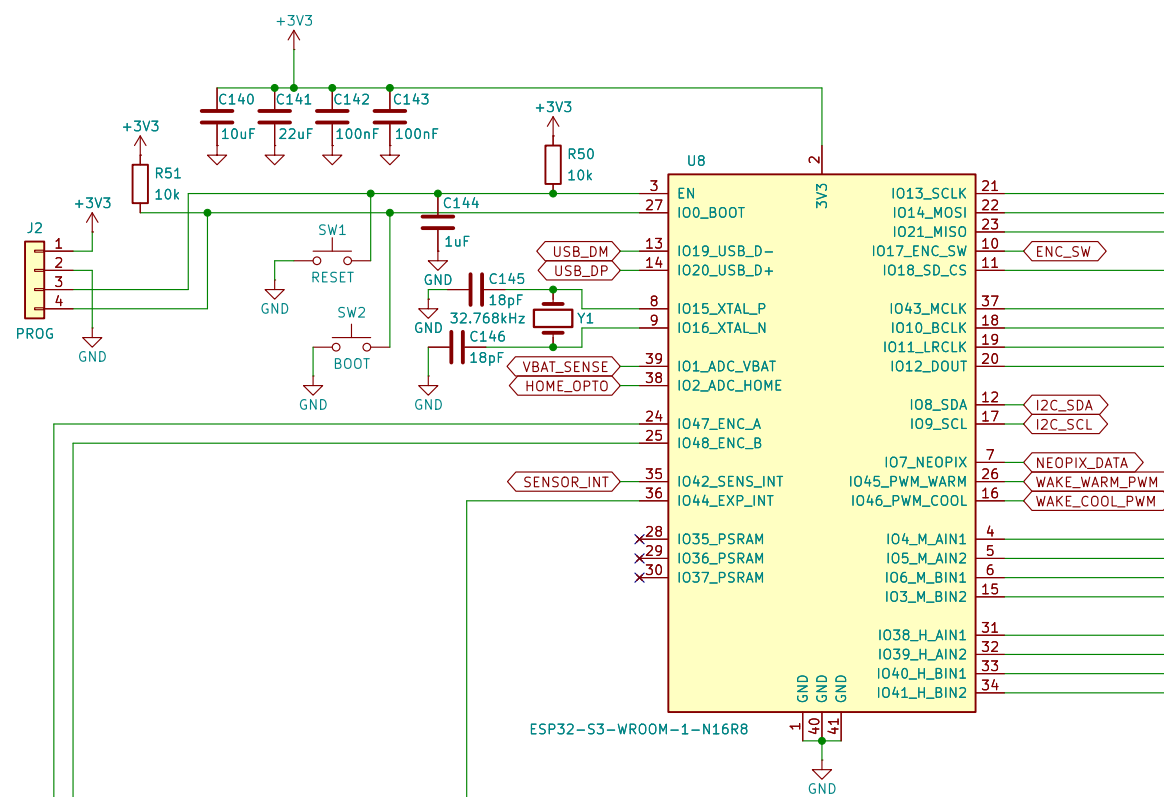
## DIAL NEOPIXEL HEAD – 2x on-PCB + 5x off-board via J12 (5 V)



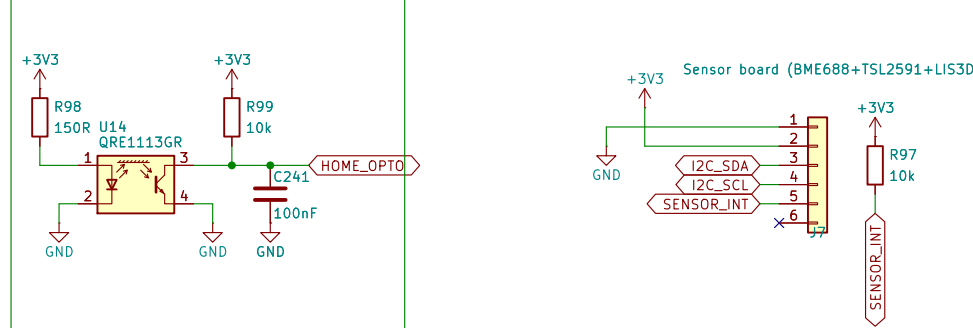
Layout: D40/D41 = dial wash either side of the shaft. C230/231 one at each pixel VDD; C237 bulk + R110 + U15 at the chain head. J12 feeds 5 more SK6812 wired off-board in series (chain pos 3-7), same rail.

## MCU – ESP32-S3-WROOM-1-N16R8 (Wi-Fi + BLE, native USB-JTAG)

ID35-37 reserved by the actual PSRAM (N16R8). Straps: ID00 high; ID45/46 low via their wake gate nets + Internal PD; ID3 Internal. Flash+console+JTAG = USB-CDC (019/20).

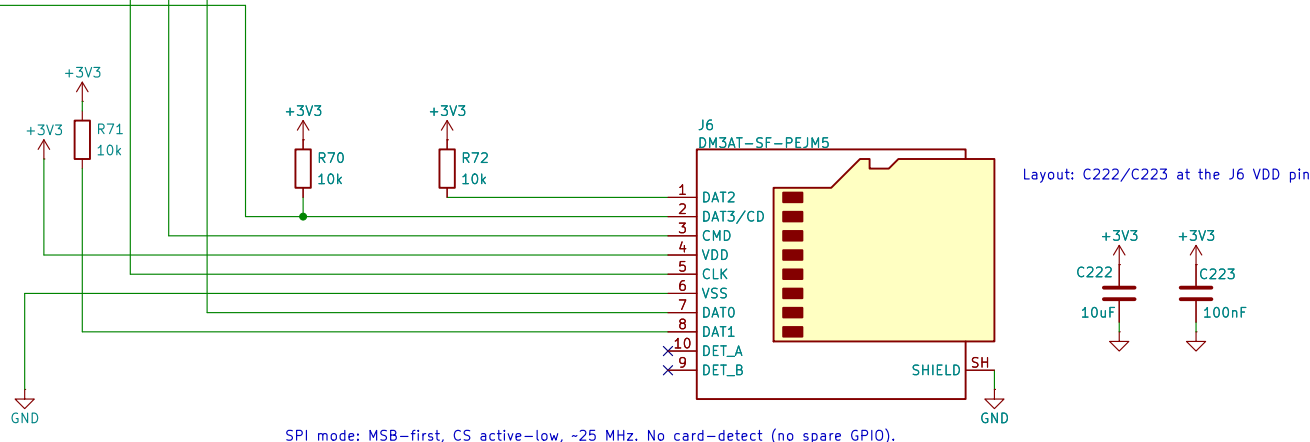


## SENSORS (I2C off-board) + HAND-HOMING (QRE1113)



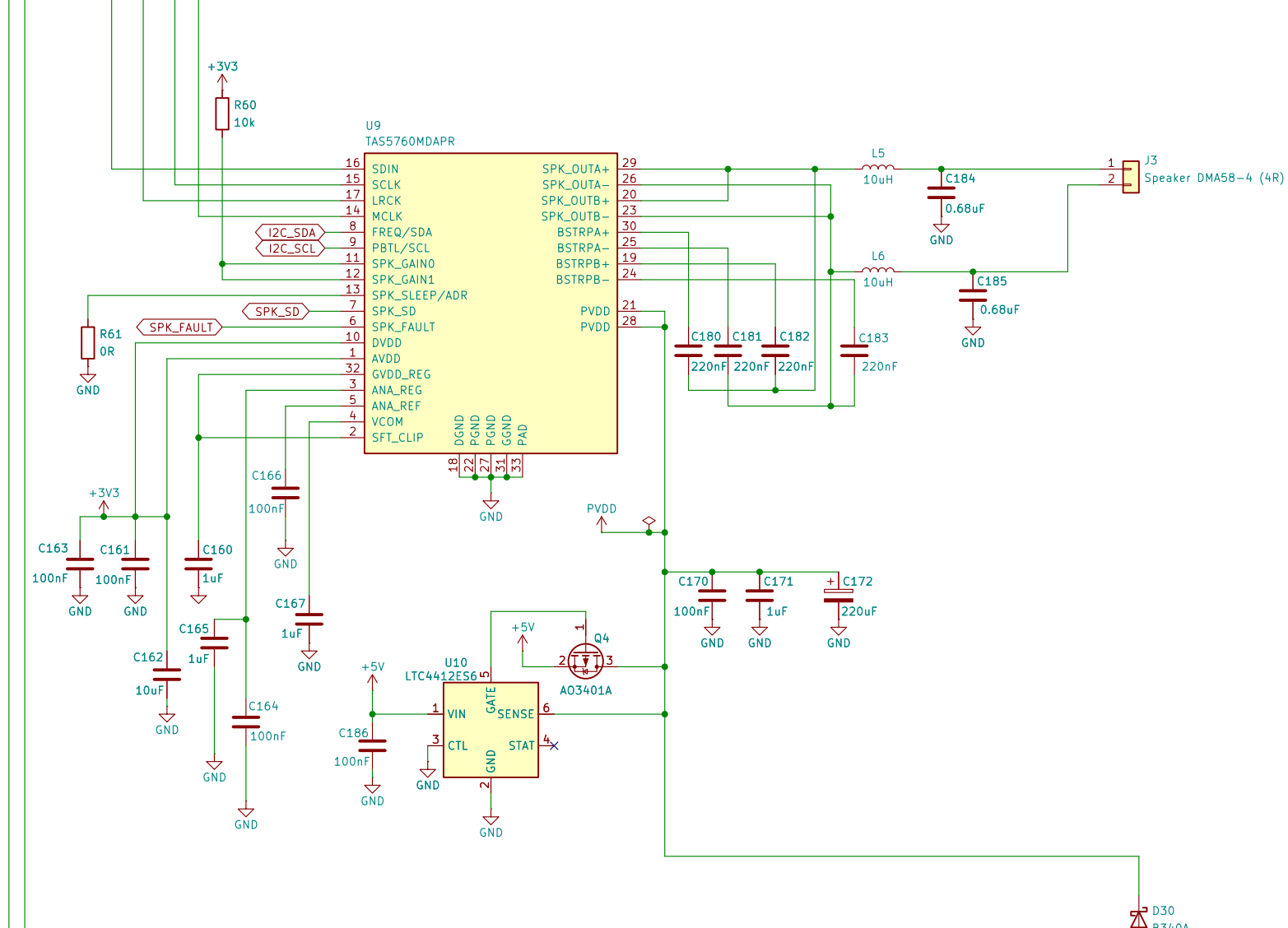
QRE1113GR faces reflective tabs on the hand hubs (no magnets).  
Sensor breakouts carry their own I2C pull-ups.

## STORAGE – microSD (DM3AT), SPI2 dedicated



SPI mode: MSB-first. CS active-low. ~25 Mhz. No card-detect (no spare GPIO).

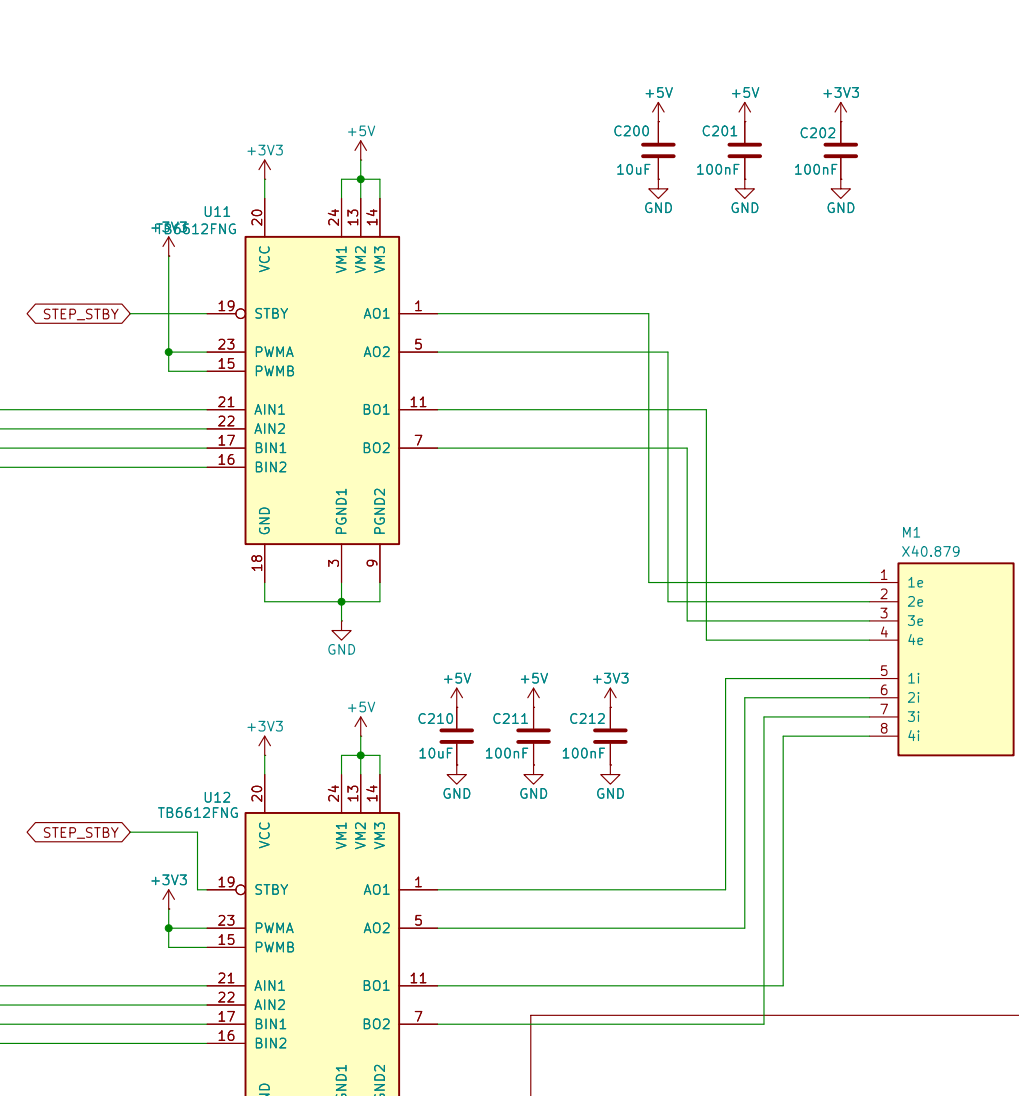
## AUDIO – TAS5760M PBTL + LC filter + PVDD mux (12V/5V)



PVDD auto-mux: 12 V wins when plugged; on battery Q4 ideal-diodes the 5 V rail -> -3 W quieter alarm. Firmware: LEDs+audio <= -12 W.  
PBTL mono (reg 0x06[7]=1): OUTA[1]OUTB. MCLK 256fs ~12.288 MHz.

## MOTOR – 2x TB6612FNG -> X40.879 dual-shaft stepper

Layout: each driver's 100uF+100nF go at its VM pins, the 100nF at its VCC pin.



PWM-on-in Microstepping: MCPWM waveforms on AIN/AIN, PWM/A/B tied high.  
STEP\_STBY (expander GPA1) low at boot -> drivers off. Coil map: esp32.md.

Tibo Mauron  
Sheet: 7  
File: clock.kicad\_sch  
Title: Wooden Clock – main board  
Size: A1  
Date: 2026-07-14  
Rev: 0.1  
KICAD E.D.A. 10.0.0  
1 of 1